



PDF Accessibility: Tools and Challenges

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Abstract. This paper presents a comparison of different post-processing software tools to make PDF documents accessible. The comparison is based on different qualitative and verifiable criteria. In addition, this paper gives an overview of different standards of PDF accessibility for people with visual impairments using screen-reader software. PDF formats, potential barriers for visually impaired users, PDF accessibility standards, and available software solutions are described in more detail. A discussion including challenges for further work is presented at the end of the paper.

Keywords: PDF accessibility · Visually impaired users · Screen readers
Tagged PDF

1 Introduction

While a plain piece of text might be perfectly accessible as such, the situation becomes much more complex as soon as a document gets larger. Navigation soon becomes an important aspect, as visually-impaired users cannot simply scan through lots of pages to get a sense of orientation. Good structure is essential, with proper headings serving as identifiers for different parts of the document.

Additionally, purely visual elements such as illustrations, drawings and pictures need an alternative textual description such that assistive technologies, like screen readers, can read out a description of the graphical content. Furthermore, complex layouts might need additional meta-information about the flow of text to ensure that assistive technologies can process content in its logical order. The various document formats have their own, although often similar, ways to add such metadata to documents.

PDF has become the de-facto standard for fixed-format document exchange and publishing. It is partially based on PostScript used for printing. PDF files are therefore more like a set of instructions on how to visually produce a given page, rather than a structured representation of content, such as HTML. Its contents are inherently less accessible due to the missing structure.

To alleviate this issue, version 1.4 of the standard ISO 32000-1 introduced “tagged PDF,” adding structural metadata. Sequences of rendering commands can be assigned a tag describing its logical significance. For example, a command drawing a line of text can have a tag marking it as a heading. Embedded images and line drawings can be

tagged as figures. Tags also can contain additional attributes, such as an alternative textual description.

All tags themselves can be structured hierarchically, forming the tag tree of a document. Aside from the hierarchical structure of elements, this tree also specifies the order in which elements should be read. This order enables assistive technologies to process content in its logical, semantically correct order instead of the order in which the rendering commands appear in the file.

2 PDF Accessibility Guidelines and Standards

The World Wide Web Consortium (W3C) publishes the Web Content Accessibility Guidelines (WCAG) currently in its second version. These guidelines are based on four basic principles for accessibility: perceivable, operable, understandable and robust. While the guidelines themselves are focused on web content (possibly represented as HTML), a subset of them also applies to documents in general.

ISO standard 14289 [1], also known as PDF/UA, defines further requirements for accessible PDF files and complements WCAG 2.1. The standard does not define new extensions to the PDF standard, but merely defines how existing facilities should be used to provide accessible documents. It puts forward requirements for processors, assistive technologies, and PDF files. The standard also outlines detailed requirements for different types of elements (headings, tables, lists, formulas, embedded content, etc.).

The Matterhorn Protocol outlines a testing model to ensure a document's accessibility in regards to PDF/UA. It consists of 136 failure conditions, of which 89 are testable by software. The conditions are organized in 31 checkpoints, categorizing the different aspects of a document covered by PDF/UA requirements. It's important to keep in mind that while the majority of conditions in the Matterhorn Protocol are machine-testable, the most critical aspects of accessibility – such as the correct structuring and proper reading order – still have to be verified manually. This process usually also requires the human verifier to be able to see the document in order to make these judgments and thus is an inherently inaccessible task.

3 Producing Accessible PDF Documents

Since accessible PDF documents are mainly based on correct tagging, tag structures should be included where possible when authoring a document. Some popular authoring software incorporates PDF/UA support. Major examples include: Adobe Acrobat [2], Microsoft Office, LibreOffice [3], and CommonLook [4].

Alternatively, an existing PDF file can be enhanced for accessibility with a suitable post-processing tool after its initial creation. Examples of such tools include Adobe Acrobat, PDFfix [5], axesPDF QuickFix [6], CommonLook, and PAVE [7, 8]. The following section describes a comparison of these PDF accessibility post-processing tools, based on verifiable ad-hoc criteria (Table 1). This list may not be complete.

Table 1. Comparison of post-processing tools for PDF accessibility

Tool name	Free of charge	Open-source	Multilingual	Web-based	Creates accessibility report	Manual tag manipulation available
Adobe Acrobat	No	No	Yes (many languages)	No (installation needed)	Yes	Yes
PDFFix	Yes	No	No (English only)	Yes	No	No
axesPDF QuickFix	No	No	Yes (English, German)	No (installation needed)	Yes	Partially
CommonLook	No	No	No	No	Yes	Yes
PAVE	Yes	Yes	Yes (English, German)	Yes	Yes	Yes

The first four criteria above describe how widely available a tool is to users worldwide: a free or low-cost tool can obviously be accessed by more users, as can a tool available in multiple languages. Being open-source allows users to adapt and expand a tool according to different needs, while web-based tools eliminate the need for users to install specialized software onto a given device. The next criterion refers to accessibility reports, which enable communication of accessibility issues with different stakeholders involved in the creation of the PDF document. Finally, many aspects of PDF accessibility are not possible without human interaction, such as adding alternative text for an image. As such, the option of manual manipulation is important, as it provides added flexibility for users when amending documents.

This comparison does not describe all important aspects of PDF post-processing tools. A highly important aspect, for example, is ease-of-use for non-experts, as well as handling large PDF files in an efficient way. However, these features are beyond the scope of this research. For web-based tools, it is also important that they provide mechanisms to protect the privacy and security of the content uploaded to their platforms. It is not known to which degree privacy and security are addressed by all tools. Additionally, none of the presented tools provides mechanisms to interpret and describe mathematical formulas and scientific graphs such that they can later be presented to a screen reader user.

4 Discussion and Future Challenges

This paper presented an overview of different aspects of PDF accessibility, as well as a comparison of different PDF accessibility post-processing tools, based on verifiable criteria.

There are several different tools available nowadays to produce accessible PDFs that comply with the corresponding standards. There are still some issues, however, regarding scientific PDF documents containing mathematical formulas or scientific graphs. Although considerable research work has been done in recognizing

mathematical formulas from PDFs or images [9–11], and there are some research prototypes available that are able to recognize mathematical formulas in PDFs or image files [12, 13], there is still no easy way to make scientific PDF documents accessible. One possibility to make scientific graphics (such as charts, graphs, and maps) accessible in PDFs is to manually write an alternative text that explains the content of the graphic. However, this is time-consuming and not pragmatic. Hence, future research and development must be done in order to make scientific PDF documents accessible. To that end, the recognition of mathematical formulas in PDF documents has to be further improved and integrated into a tool such as PAVE, and an intuitive way to generate automated alt-text explaining the contents of graphs and formulas must be found. This would allow millions of technical and scientific authors to easily create PDF documents that are accessible to readers with visual impairments, which in our opinion is well worth the effort.

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